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AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Tourism & Hospitality Booking Dashboard and Revenue Insights

Project Domain

Excel with Copilot

A Project Report

Submitted in partial fulfillment of the requirements of the AIWC Project

by: Priyanka Kumari

Student Name	Student Email	Guide
Priyanka Kumari	**p0385754@gmail.com**	**Umamaheswari**

Institution:

Brainware University

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ABSTRACT OF THE PROJECT

The tourism and hospitality industry generates a large amount of booking and revenue data from hotel reservations, customer records, and payment transactions. However, raw data is often difficult to interpret without proper analysis and visualization.

This project, *"Tourism & Hospitality Booking Dashboard and Revenue Insights,"* focuses on analyzing hotel booking data using Microsoft Excel with Copilot assistance. The dataset obtained from Kaggle was cleaned and organized by removing duplicates, handling missing values, and correcting data formats to ensure accurate analysis.

An interactive dashboard was created using charts, pivot tables, and key performance indicators to visualize booking trends, revenue distribution, customer types, and payment preferences. The dashboard helps convert raw data into clear visual insights.

The results highlight important patterns in bookings and revenue, which can help hospitality businesses make better decisions related to pricing, marketing strategies, and resource management.

TABLE OF CONTENTS

Acknowledgement

Abstract of the Project

Chapter 1 – Introduction

Chapter 2 - Literature Survey

Chapter 3 – Proposed Methodology

Chapter 4 – Implementation and Result

Chapter 5 – Discussions and Conclusions

References

List of Figures

Figure No.	Figure Titles	Page No.
1	Uncleaned Dataset Screenshot	15
2	Cleaned Data Screenshot	16
3	Revenue by Hotel	17
4	Revenue by Hotel and Room Type	18
5	Revenue by seasons	18
6	Sum of Rooms Booked by Rooms Available	19
7	Revenue by Payment Methods	19
8	Sum of Rooms booked in different Seasons	20
9	Data Flow Diagram	12

List of Tables

Table No.	Table Name	Page No.
1	Hardware Requirements	13
2	Software Requirements	13

Chapter 1

INTRODUCTION

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and Al-assisted Reporting.

1.1 Problem Statement

The tourism and hospitality industry generates large volumes of booking and revenue data from multiple sources.
This data is often unorganized and difficult to interpret, making analysis challenging.
Lack of proper visualization makes it hard to identify booking trends, customer behaviour, and revenue patterns.
This impacts decision-making in pricing, marketing, and resource management.
Existing methods like basic spreadsheets and manual analysis are time-consuming and ineffective.
There is a need for a structured and efficient approach to clean, analyze, and visualize data.
This project addresses the problem using Microsoft Excel with Copilot assistance to create an interactive dashboard for better insights.

1.2 Objectives

- To analyze tourism and hospitality booking data to identify key trends in bookings, revenue, and customer behavior.
- To clean and organize raw data for accurate and meaningful analysis.
- To create an interactive dashboard using Microsoft Excel for easy visualization of insights.
- To understand factors affecting revenue such as room types, seasons, and customer preferences.
- To support better decision-making in pricing, marketing, and resource management.
- To use Copilot Agent to enhance data analysis and generate insights efficiently.

1.3 Scope of the Project

The scope of this project is to analyze tourism and hospitality booking data and present the insights through a clear and interactive dashboard with Integration with Microsoft Copilot agent for better interactions and assistance with the cleaned data. The project focuses on cleaning

and organizing the dataset, analyzing booking patterns, revenue trends, and customer behavior. Using tools such as Microsoft Excel and a dataset obtained from Kaggle, the project aims to convert raw data into meaningful visual information. The dashboard helps in understanding key aspects such as room demand, seasonal booking trends, and overall revenue performance in the hospitality sector. However, the analysis is limited to the available dataset and does not include real-time data or advanced predictive models.

Chapter 2

LITERATURE SURVEY

1. (Taiga Saito, Akihiko Takahashi, Noriaki Koide, Yu Ichifuji – 2019) Conducted research on **Application of Online Booking Data to Hotel Revenue Management.** This study explains how online booking data can be used to improve hotel revenue management. The authors analyzed booking information collected from hotel reservation platforms to understand customer demand and booking patterns. Their research shows that analyzing online reservation data helps hotels forecast demand more accurately and adjust room prices accordingly. As a result, hotels can maximize revenue and reduce losses caused by cancellations or unsold rooms.
2. (Stanislav Ivanov, Vladimir Zhechev – 2012) Conducted reseach **on Hotel Revenue Management.** This paper reviews previous studies related to revenue management in the hospitality industry. The authors discuss how hotels use pricing strategies, demand forecasting, and inventory control to maximize profits. The research highlights that revenue management has become an essential practice for hotels to remain competitive in the tourism market. It also suggests that technology and data analysis play a significant role in improving decision-making in hotel management.
3. (Chatarin Subying, Chanin Yoopetch – 2023) Conducted research on **A Bibliometric Review of Revenue Management in the Tourism and Hospitality Industry (1989–2021).** This study provides an overview of research conducted on revenue management in tourism and hospitality over more than three decades. The authors examined many academic publications to identify trends, research areas, and developments in this field. Their findings show that topics such as dynamic pricing, demand forecasting, and customer behavior analysis have gained significant attention in recent years. The study also highlights the growing use of big data and analytics in hospitality research.

2.1 Existing Work and Research

Several studies have explored the use of data analytics and revenue management in the tourism and hospitality industry. Research by Taiga Saito et al. (2019) highlighted the importance of analyzing online booking data to improve hotel revenue management. Their study shows that booking data can help hotels forecast customer demand and adjust pricing strategies to maximize revenue.

Another study conducted by Stanislav Ivanov and Vladimir Zhechev (2012) reviewed various revenue management techniques used in hotels. The research emphasized that demand forecasting, dynamic pricing, and inventory control are essential for improving hotel profitability and managing room availability efficiently.

Research by Rimo Das, Harshinder Chadha, and Somnath Banerjee (2021) introduced forecasting models that analyze market trends and booking patterns. Their findings suggest that predictive models can help hospitality businesses anticipate customer demand and optimize pricing decisions.

Another study by Luis Sánchez-Medina and colleagues (2021) examined factors that influence hotel room pricing using large datasets. The study concluded that variables such as hotel category, location, customer ratings, and amenities significantly affect booking decisions and revenue generation.

Although these studies provide valuable insights into revenue management and booking analytics, many of them focus mainly on theoretical models and advanced analytical systems. There is still a need for simple and accessible dashboard solutions that can help businesses easily analyze booking data and visualize key performance indicators.

The present project addresses this gap by developing a Tourism & Hospitality Booking Dashboard that analyses booking data, identifies trends, and provides revenue insights using easy-to-use data visualization techniques.

2.2 Tools and Technologies Review

The following tools and technologies are used to develop the Tourism & Hospitality Booking Dashboard:

1. Microsoft Excel

Microsoft Excel is widely used for data analysis, cleaning, and visualization. It provides features such as pivot tables, charts, conditional formatting, and formulas that help in transforming raw booking data into meaningful insights. Excel also allows the creation of interactive dashboards to monitor key performance indicators.

2. Copilot

Copilot Agent assists in generating insights, explanations, and analytical interpretations from datasets. It helps users understand patterns and trends in data more effectively.
This tool simplifies the process of data cleaning, analysis, visualization, and dashboard creation, making it easier to generate meaningful insights from tourism and hospitality booking data.

Chapter 3

PROPOSED METHODOLOGY

This chapter explains the methodology used to achieve the objectives of the project Tourism & Hospitality Booking Dashboard and Revenue Insights. It describes the system design, data flow, data processing techniques, and the tools used to generate meaningful insights from booking data and transferring the cleaned data to copilot for assistance in interpretation.

3.1 System Design

Data Source (Raw Data) : The dataset used in this project was obtained from Kaggle, an online platform that provides publicly available datasets for data analysis and machine learning projects. The dataset contains information related to tourism and hospitality bookings, including details such as hotel name, booking date, room type, number of guests, booking status, and revenue generated.

Data Cleaning : The dataset was cleaned using Microsoft Excel by performing the following steps:

1. **Removing Duplicate Records**
   Duplicate rows in the dataset were identified and removed using Excel's *Remove Duplicates* feature to ensure each booking record appeared only once.
2. **Handling Missing Values**
   Some fields in the dataset contained blank or null values. These were handled by either filling appropriate values where possible or removing incomplete records that could affect the analysis.
3. **Correcting Data Formats**
   Data formats such as dates, numbers, and text entries were standardized. For example, booking dates were converted into proper date format and revenue values were formatted as numbers.
4. **Filtering Irrelevant Data**
   Unnecessary columns or irrelevant data fields that were not required for the analysis were removed to simplify the dataset.
5. **Ensuring Data Consistency**
   Inconsistent entries such as spelling variations or category mismatches were corrected to maintain uniformity in the dataset.
   After completing these cleaning steps, the dataset became structured and reliable for further analysis and visualization in the dashboard.

Analysis : The following techniques were used during the analysis process:

1. **Pivot Table Analysis**
   Pivot tables were created to summarize large amounts of booking data. This helped in identifying total bookings, revenue generated, and category-wise performance such as room types or hotel categories.

2. **Revenue and Booking Trend Analysis**
   Booking data was analyzed to identify trends such as peak booking periods, seasonal demand, and revenue generation patterns. This helped in understanding which periods generated the highest bookings and revenue.

3. **Key Performance Indicator (KPI) Identification**
   Important KPIs such as total bookings, revenue generated, and category-wise performance were identified. These KPIs were later used in the dashboard to present insights clearly.

4. **Visualization :** The final output is an interactive dashboard and summary report created using Microsoft Excel with Copilot Agent assistance. It presents key metrics such as total bookings, revenue trends, customer types, and payment preferences using charts, pivot tables, and filters for easy analysis. The dashboard highlights important insights from the cleaned dataset, helping stakeholders understand booking patterns and make better decisions in pricing, marketing, and resource management.

3.2 Components Used

The project uses several components to perform data analysis and create the tourism and hospitality booking dashboard. These components help in data processing, visualization, and generating insights.

1. Microsoft Excel

Microsoft Excel is the main tool used in this project for data cleaning, analysis, and visualization. It provides features such as pivot tables, formulas, charts, and dashboard creation.

2. Kaggle Dataset

The dataset used in this project was obtained from Kaggle. It provides tourism and hospitality booking data including hotel details, booking information, revenue, and customer data.

3. Data Analysis Tools
Various Excel features such as sorting, filtering, pivot tables, and formulas were used to analyze booking trends, revenue patterns, and customer behavior.

4. Visualization Tools
Charts, graphs, and KPI indicators were used to create an interactive dashboard that clearly presents booking statistics and revenue insights.

5. Copilot Tools

Copilot Agent was used on the cleaned dataset to analyze data and identify key trends and insights.
It also assisted in generating summaries and improving dashboard visualizations efficiently.

3.3 Data Flow Diagram (DFD)

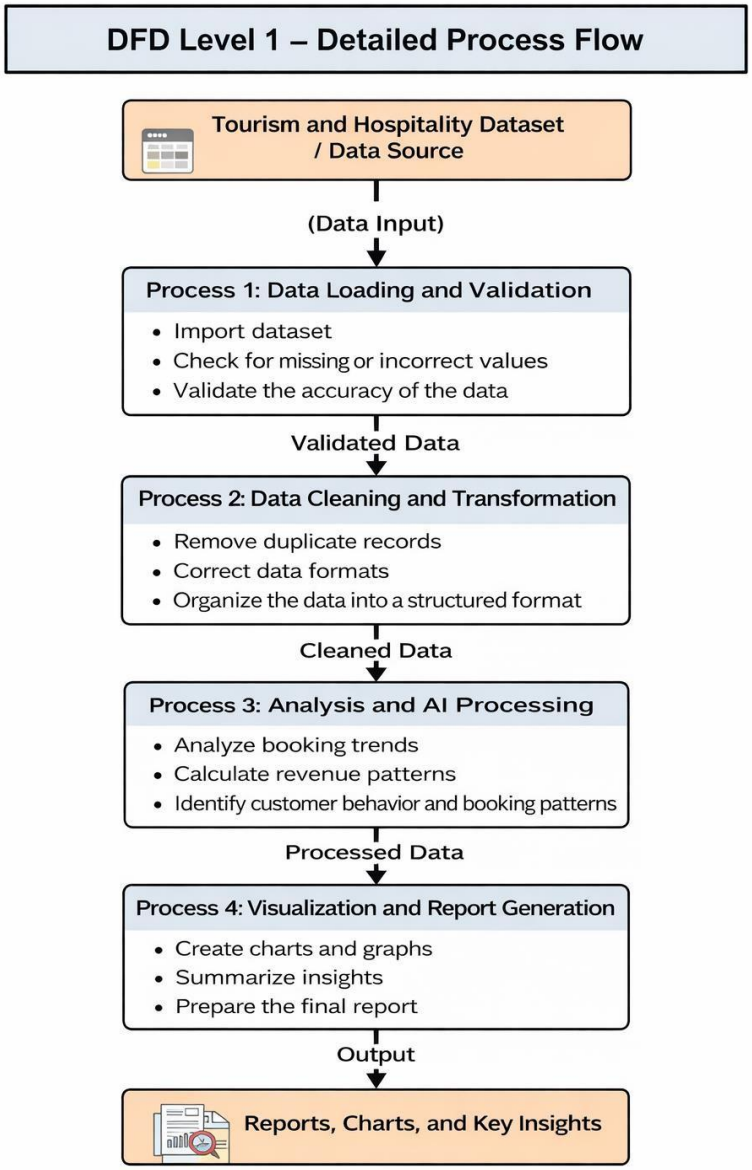

Fig. 9

3.4 Advantages

- **Better Data Understanding**
  The dashboard helps convert raw tourism and hospitality booking data into clear visual insights, making it easier to understand patterns and trends.

- **Improved Decision Making**
  By analyzing bookings, revenue, and customer behavior, businesses can make better decisions related to pricing, marketing strategies, and resource allocation.
- **Easy Data Visualization**
  Using Microsoft Excel, complex data is presented through charts, graphs, and tables, which simplifies the interpretation of information.
- **Identification of Trends**
  The dashboard helps identify seasonal booking patterns, popular room types, and revenue distribution, which are useful for planning business strategies.
- **Time Efficient Analysis**
  Instead of manually analyzing large datasets, the dashboard provides quick insights through automated calculations and visual summaries.

3.5 Requirement Specification

Hardware Requirements

Components	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 × 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

Chapter 4

IMPLEMENTATION AND RESULT

This chapter presents the implementation of the project and the results obtained after performing data cleaning, analysis, and visualization. It explains how the raw dataset was processed and how the final dashboard was created to generate insights from tourism and hospitality booking data and further cleaned data was integrated to copilot agent for better interaction and assistance.

5.1 Data Cleaning Results

Dataset Source

The dataset used in this project was downloaded from Kaggle. It contains tourism and hospitality booking data including hotel name, room type, booking date, customer type, revenue, booking status, payment mode, and other related details.

Original Dataset Size

The original dataset consisted of multiple rows and columns containing booking and revenue information for different hotels and room categories.

Issues Found

During the initial inspection of the dataset, several issues were identified such as:

- Duplicate records in booking entries
- Missing or null values in some columns
- Inconsistent data formats for dates and numerical values
- Minor formatting inconsistencies

Cleaning Steps Applied

Data cleaning was performed using Microsoft Excel. The following steps were applied:

- Duplicate rows were identified and removed.
- Missing values were handled by correcting or removing incomplete records.
- Date and numerical formats were standardized.
- Irrelevant columns were removed to simplify the dataset.

Cleaned Dataset Size

After completing the cleaning process, the dataset became structured and ready for analysis. The cleaned dataset was then used to perform further analysis and visualization with Microsoft Copilot assistance.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Hotel Name	Room Type	Booking Date	Revenue	stomer Ty	Season	Rooms Available	oms Book	oking Stat	Booking Lead Time (Days)	Repeat Guest	Payment Mode	Stay Duration (Nights)	Discount (%)	Net Revenue	
ake View Lodge		2025/11/16	92840	Family	Off-Peak	50	4	Confirmed	15	Yes	Net Banking	2	15	78914	
ea View Resort	Standard	2025-12-17	44598	Couple	Off-Peak	50	1	Confirmed	6	No	Cash	5	10	40138.2	
ake View Lodge	Deluxe	2025/10/06	48552	Individual	Peak	50	4.0 USD	Confirmed	1	Yes	Net Banking	3	0	48552	
illtop Inn	Suite	02-02-2025	105192	Couple	Off-Peak	50	2	Confirmed	20	Yes	Cash	5	5	99932.4	
unrise Hotel		2025-02-24	199560	Family	Off-Peak	50	5	Confirmed	13	Yes	UPI	4	5	189582	
ea View Resort	Suite	2025-11-11	26958	Couple	Off-Peak	50	2	Confirmed	25	Yes	Debit Card		10	24262.2	
oyal Palace	Deluxe	2025-01-06	53788		Peak	50	4	Confirmed	15	No	Credit Card	4	15	45719.8	
ea View Resort		2025/07/22	100080	Corporate	Shoulder	50	2	Confirmed	3	No		3	10	90072	
unrise Hotel	Standard	2025-07-18	289360		Shoulder	50	5	Confirmed	5	Yes	UPI	4	10	260424	
unrise Hotel	Deluxe	2025-04-19	40080	Couple	Peak	50	2	Confirmed	15	No	Net Banking	2	10	36072	
ake View Lodge	Standard	2025-07-06	43464	Family	Peak	50	4	Confirmed	20	Yes	Credit Card	1	10	39117.6	
ity Stay	Standard	01.23.2025	245140	Family		50	5	Confirmed	27	Yes	UPI	4	5		
illtop Inn	Suite	2025/12/13	55576	Couple	Off-Peak	50	1	Confirmed	20	Yes	Debit Card	4	10	50018.4	
unrise Hotel	Deluxe	2025/10/02	51595	Couple	Off-Peak	50	5	Confirmed	20	Yes	Cash	2	10	46435.5	
oyal Palace	Deluxe	2025/05/22	0	Couple	Peak	50	1	Cancelled	20	No	Net Banking	4	0	0	
ity Stay	Suite	15 Aug 2025	0	Corporate	Shoulder		2	Cancelled	26	No	Credit Card	2	5	0	
illtop Inn	Suite	19 Feb 2025	16980	Couple	Off-Peak	50	3	Confirmed	18	No	UPI	1	15	14433	
oyal Palace	Deluxe		141732	Corporate	Off-Peak	50		Confirmed	11	No	Cash	2	15	120472.2	
ity Stay	Deluxe	2025-03-27	32394	Individual	Off-Peak	50	1	Confirmed	21	Yes	Cash	3	5	30774.3	
one	Suite	2025-12-16	22328	Corporate	Off-Peak	50	4.0 USD	Confirmed	13	Yes	Credit Card	2	10	20095.2	
oyal Palace	Deluxe	2025/06/21	10298	Couple	Peak	50	2	Confirmed	25	No	Net Banking	3	15	8753.3 USD	
unrise Hotel	Deluxe	2025/09/22	199072	Couple	Off-Peak	50	4	Confirmed	16	Yes	UPI	1	15	169211.2	
ity Stay	Suite	2025/11/05	8260	Corporate		50	2	Confirmed	23	Yes	Cash	5	10		
ake View Lodge	Standard	2025-02-21	26682	Individual	Off-Peak	50	2	Confirmed	14	No	Debit Card	4	15	22679.7	

Fig. 1 (Uncleaned Dataset)

Hotel Name	Room Type	Booking Date	Revenue	Customer Type	Season	Rooms Available	Rooms Booked	Booking Status	Booking Lead Time (Days)	Repeat Guest	Payment Mod
Lake View Lodge	Deluxe	16/11/2025	92840	Family	Off-Peak	50	4	Confirmed	15	Yes	Net Banking
Sea View Resort	Standard	17/12/2025	44598	Couple	Off-Peak	50	1	Confirmed	6	No	Cash
Lake View Lodge	Deluxe	10/06/2025	48552	Individual	Peak	50	4	Confirmed	1	Yes	Net Banking
Hilltop Inn	Suite	02/02/2025	105192	Couple	Off-Peak	50	2	Confirmed	20	Yes	Cash
Sunrise Hotel	Suite	24/02/2025	199560	Family	Off-Peak	50	5	Confirmed	13	Yes	UPI
Sea View Resort	Suite	11/11/2025	26958	Couple	Off-Peak	50	2	Confirmed	25	Yes	Debit Card
Royal Palace	Deluxe	01/06/2025	53788	Family	Peak	50	4	Confirmed	15	No	Credit Card
Sea View Resort	Suite	22/07/2025	100080	Corporate	Shoulder	50	2	Confirmed	3	No	Debit Card
Sunrise Hotel	Standard	18/07/2025	289360	Family	Shoulder	50	5	Confirmed	5	Yes	UPI
Sunrise Hotel	Deluxe	19/04/2025	40080	Couple	Peak	50	2	Confirmed	15	No	Net Banking
Lake View Lodge	Standard	07/06/2025	43464	Family	Peak	50	4	Confirmed	20	Yes	Credit Card
City Stay	Standard	23/01/2025	245140	Family	Off-Peak	50	5	Confirmed	27	Yes	UPI
Hilltop Inn	Suite	13/12/2025	55576	Couple	Off-Peak	50	1	Confirmed	20	Yes	Debit Card
Sunrise Hotel	Deluxe	10/02/2025	51595	Couple	Off-Peak	50	5	Confirmed	20	Yes	Cash
Royal Palace	Deluxe	22/05/2025	0	Couple	Peak	50	1	Cancelled	20	No	Net Banking
City Stay	Suite	15/08/2025	0	Corporate	Shoulder	50	2	Cancelled	26	No	Credit Card
Hilltop Inn	Suite	19/02/2025	16980	Couple	Off-Peak	50	3	Confirmed	18	No	UPI
Royal Palace	Deluxe	06/10/2025	141732	Corporate	Off-Peak	50	4	Confirmed	11	No	Cash
City Stay	Deluxe	27/03/2025	32394	Individual	Off-Peak	50	1	Confirmed	21	Yes	Cash
Royal Palace	Suite	16/12/2025	22328	Corporate	Off-Peak	50	4	Confirmed	13	Yes	Credit Card
Royal Palace	Deluxe	21/06/2025	10298	Couple	Peak	50	2	Confirmed	25	No	Net Banking

Fig. 2 (Cleaned Dataset)

5.2 Visualization

Chart 1 :

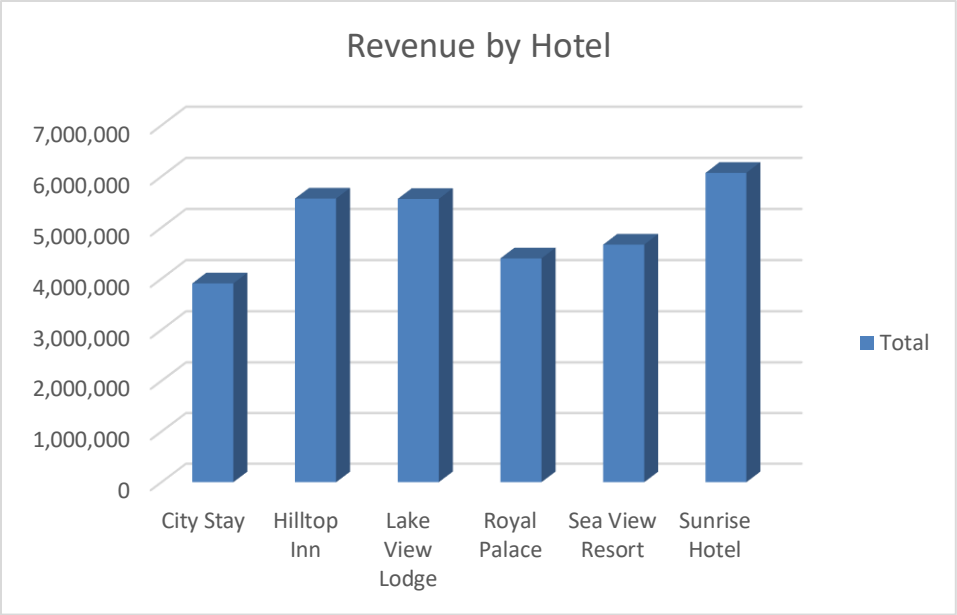

Fig. 3

This chart represents the total revenue generated by different hotels, allowing an easy comparison of their performance. It shows that Sunrise Hotel has the highest revenue, followed by Hilltop Inn and Lake View Lodge, while City Stay generates the lowest revenue. Overall, the chart helps identify top-performing and low-performing hotels, supporting better business decisions.

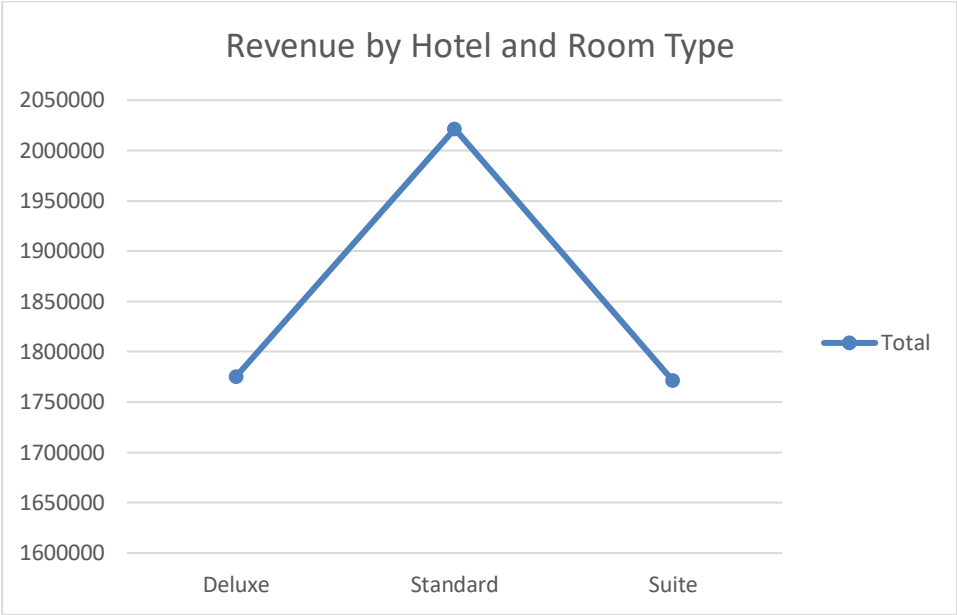

Fig. 4

This chart shows the revenue generated from different room types, including Deluxe, Standard, and Suite. It indicates that Standard rooms generate the highest revenue compared to Deluxe and Suite rooms, which have almost similar but lower revenue. The chart helps in understanding which room type contributes most to overall revenue, supporting better pricing and room allocation decisions.

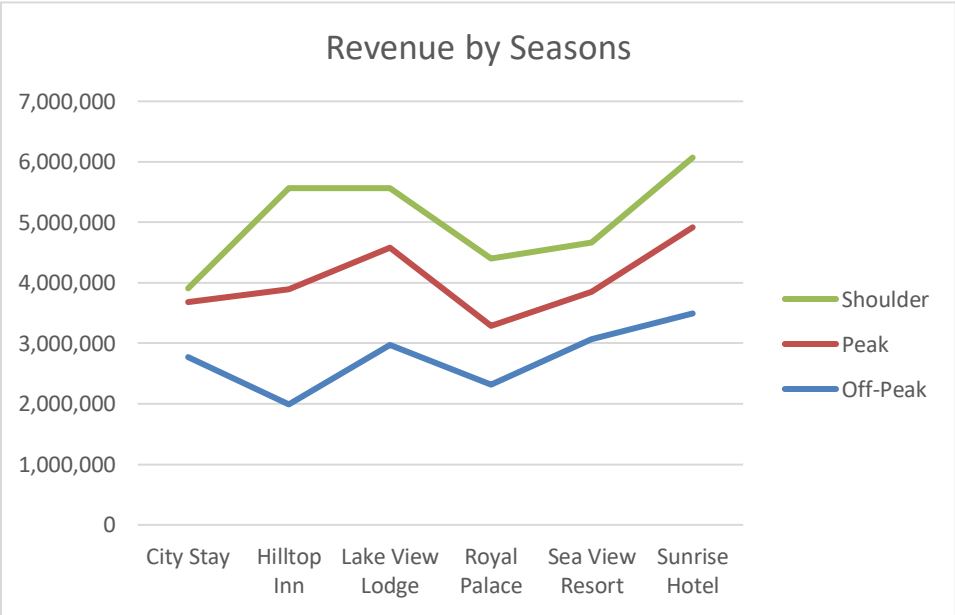

Fig. 5

This chart shows how revenue varies across different seasons such as Peak, Shoulder, and Off-Peak for each hotel. It indicates that revenue is highest during the Peak season, followed by Shoulder season , while Off-Peak generates the lowest revenue. This helps businesses understand seasonal demand and plan pricing and marketing strategies accordingly.

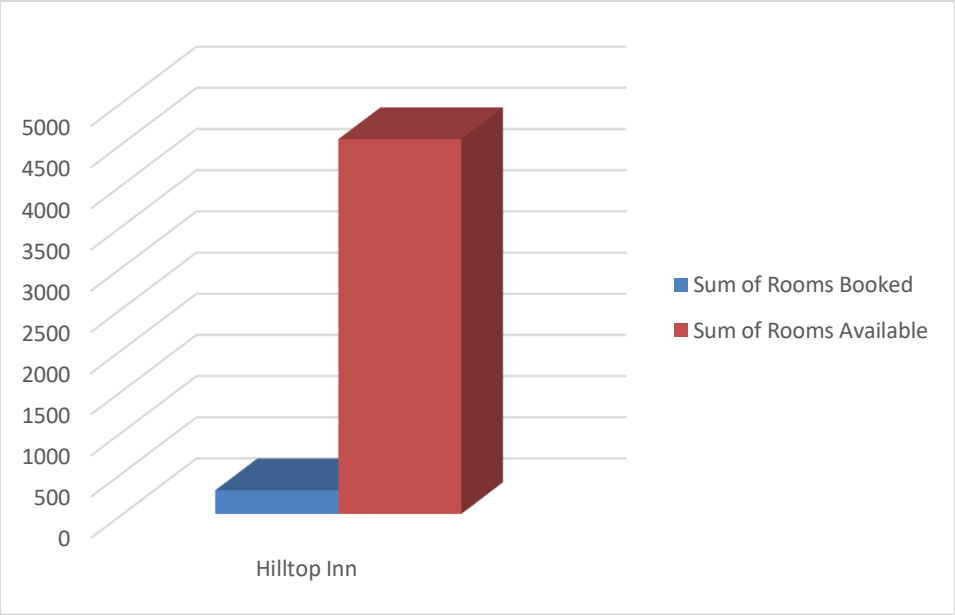

Fig. 6

This chart compares the number of rooms booked with the total rooms available for a hotel. It shows that only a portion of available rooms are booked, helping identify occupancy levels and efficiency in room utilization. This insight supports better inventory and capacity planning.

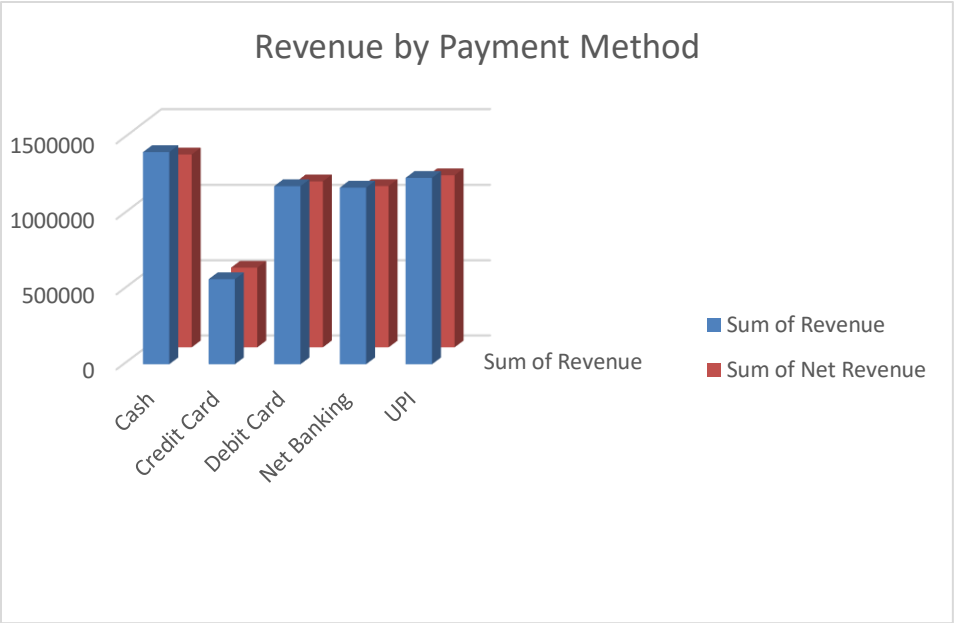

Fig. 7

This chart displays revenue generated through different payment methods such as Cash, Credit Card, Debit Card, Net Banking, and UPI. It helps identify the most preferred payment options among customers, allowing businesses to optimize their payment systems and improve customer convenience.

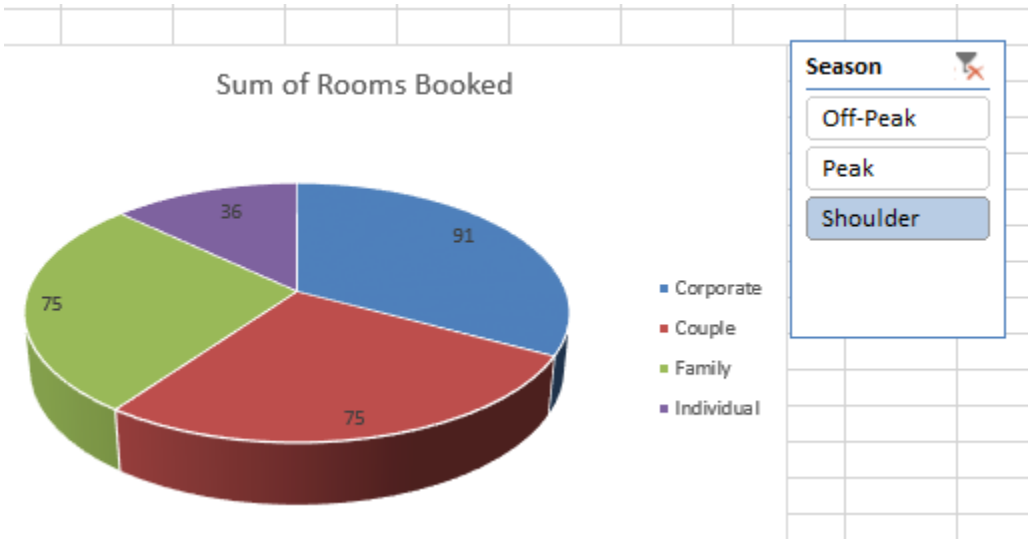

Fig. 8

The image shows pie chart titled "Sum of Rooms Booked," which displays how hotel bookings are distributed among four customer types during the selected **Shoulder** season. Corporate bookings make up the largest share with 91 rooms, followed by couples and families with 75 rooms each, while individual travelers account for the smallest portion at 36 rooms. A slicer on the right allows switching between Off-Peak, Peak, and Shoulder seasons, with Shoulder currently selected, indicating that the chart reflects data for that specific period.

Chapter 5

DISCUSSIONS AND CONCLUSIONS

5.1 Key Findings

- The dataset includes 500 hotel booking records across 6 hotels, enabling analysis of booking trends, revenue, and customer behavior.
- Sunrise Hotel is the top performer in terms of revenue, indicating strong demand.
- Suite rooms are the most preferred, showing higher interest in premium accommodation.
- Booking confirmation rate is high, with most bookings successfully completed.
- Off-Peak season shows the highest bookings, indicating consistent demand throughout the year.
- Discounts play an important role in influencing customer bookings.
- Higher revenue is driven by premium rooms, longer stays, and repeat customers.

5.2 Limitations

- **Limited Dataset Size**
  The analysis is based on a dataset obtained from Kaggle, which may not represent the complete real-world tourism and hospitality industry.
- **Static Data Source**
  The dataset used in the project is static and does not update automatically. Therefore, the dashboard does not reflect real-time booking or revenue information.
- **Limited Analytical Tools**
  The dashboard was created using Microsoft Excel, which has some limitations compared to advanced analytics tools used in large-scale hospitality businesses.
- **Simplified Analysis**
  The analysis mainly focuses on booking trends, revenue insights, and customer patterns. More complex factors such as market competition, pricing strategies, and external economic conditions were not included.
- **Data Accuracy Dependence**
  The insights generated from the dashboard depend on the accuracy and

completeness of the dataset. Any errors in the data may affect the reliability of the results.

5.3 Conclusions

This study examined tourism and hospitality booking data to understand booking behavior, revenue performance, and customer patterns within the hospitality sector. The dataset obtained from Kaggle was cleaned, organized, and analyzed using Microsoft Excel to transform raw information into meaningful insights with Copilot Assistace. Through the use of charts, pivot tables, and an interactive dashboard, the study was able to highlight key trends related to room demand, seasonal booking variations, and revenue distribution.

The findings suggest that data visualization tools can significantly improve the interpretation of booking data by presenting complex information in a clear and structured format. The dashboard developed in this project provides a simplified overview of important performance indicators such as booking trends, customer behavior, and revenue generation. These insights can assist hospitality businesses in making more informed decisions regarding marketing strategies, pricing, and resource allocation.

Overall, the study demonstrates the value of using analytical dashboards to support data-driven decision making in the tourism and hospitality industry. By converting raw data into actionable insights, such systems can help organizations better understand market trends and improve operational performance.

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